

Programming a 2D Ising Model/Determining the heat capacity

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Calculating the heat capacity

As we have seen, increasing the temperature above T_C induces a phase transition — the magnetisation of the system rapidly drops, but it can be hard to use this information to pinpoint the Curie temperature itself. As well as demonstrating the closed form solution to the partition function that we mentioned in the introduction, Lars Onsager also demonstrated that the heat capacity of the 2D Ising model should become very strongly peaked at the phase transition temperature (in fact, when $T = T_C$ exactly, C diverges).

TASK 7a:

By definition, $C = \frac{\partial \langle E \rangle}{\partial T}$. From this, show that $C = \frac{\text{Var}[E]}{k_B T^2}$, where $\text{Var}[E]$ is the variance in E .

Using this relation, and the data that you generated in the previous sections, you can now determine the heat capacity of your lattice as a function of temperature and system size. In a similar way, we can also define another quantity called the magnetic susceptibility, χ , which depends on the average properties of the order parameter of the system:

$$\chi = \beta(\langle M^2 \rangle - \langle M \rangle^2)$$

which will behave in a similar way to the heat capacity. Here $\beta = \frac{1}{k_B T}$ is the inverse of the thermodynamic temperature, often used in statistical thermodynamics.

TASK 7b:

Write a Python script to make a plot showing the heat capacity and susceptibility versus temperature for each of your lattice sizes from the previous section. You may need to do some research to recall the connection between the variance of a variable, $\text{Var}[X]$, the mean of its square $\langle X^2 \rangle$, and its squared mean $\langle X \rangle^2$. You may find that the data around the peak is very noisy — this is normal, and is a result of being in the critical region. As before, use the plot controls to save a SVG or PNG image of your plot and attach this to the report.

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